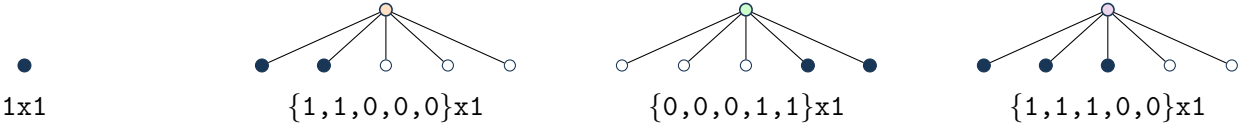


Definition and first terms

$6 A(x) = 5 + 1 x + A(x)^5$, with the origin-normalized solution.
Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+10*T^2+10*T^3+5*T^4+1*T^5$.
 $a(0), \dots, a(7) = 1, 1, 10, 210, 5505, 161601, 5082420, 167451780$.
Kernel used throughout: $\rho(u) = u(1 - 10 u^1 - 10 u^2 - 5 u^3 - 1 u^4)$, so $x = \rho(T(x))$.

Typogeometry



literal 5-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 10, Δ_3 : 10, Δ_4 : 5, Δ_5 : 1.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_4) \in \mathbb{Z}_{\geq 0}^4 : \sum_{j=1}^4 j k_j = n - 1\}, \quad K = \sum_{j=1}^4 k_j,$$
$$a(n) = \frac{1}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_4} 10^{k_1} 10^{k_2} 5^{k_3} 1^{k_4}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^4 P_r(n) a(n+r) = 0, \quad n \geq 1, \quad \text{with}$$
$$\begin{aligned} P_0(n) &= -3125 n^4 - 12500 n^3 - 13750 n^2 - 2500 n + 1155 \\ P_1(n) &= -62500 n^4 - 343750 n^3 - 668750 n^2 - 546875 n - 159375 \\ P_2(n) &= -468750 n^4 - 3281250 n^3 - 8390625 n^2 - 9234375 n - 3656250 \\ P_3(n) &= -1562500 n^4 - 13281250 n^3 - 40625000 n^2 - 52343750 n - 23437500 \\ P_4(n) &= 37531 n^4 + 375310 n^3 + 1313585 n^2 + 1876550 n + 900744 \end{aligned}$$

Differential equation

The scalar linear ODE has order 4. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n, u) = N(n, u) / \rho(u)^3, \quad \deg_u N = 16.$$

$$\text{Telescoping identity: } \sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u)).$$

Matrix dimensions: 10×10 ; remainder 4×5 , rank 4, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.